CSE655: Network Science
Analysis & Prediction of Interactions in Large Social Networks
Project Report

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April 16, 2026

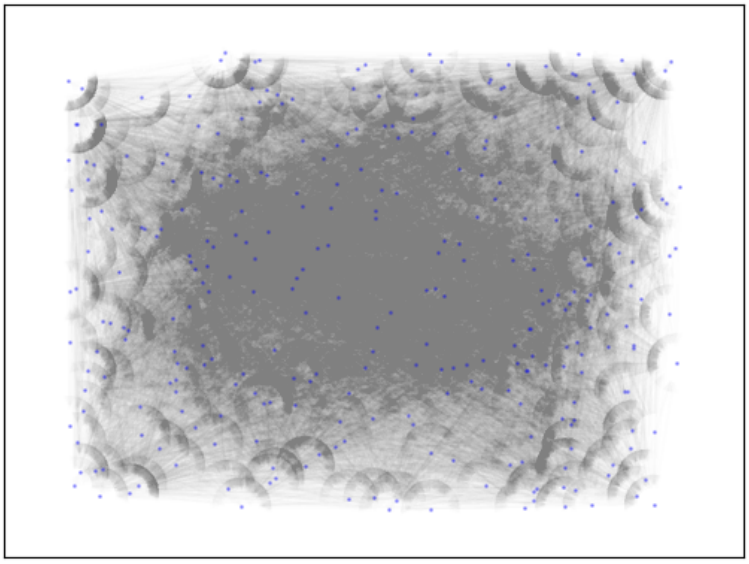
Dataset & EDA

We are using the Reddit hyperlink network. This network represents the directed connections between two subreddits (a subreddit is a community on Reddit). The network is extracted from publicly available Reddit data of 2.5 years from Jan 2014 to April 2017.
The subreddit-to-subreddit hyperlink network is extracted from the posts that create hyperlinks from one subreddit to another. We say a hyperlink originates from a post in the source community and links to a post in the target community. Each hyperlink is annotated with three properties: the timestamp, the sentiment of the source community post towards the target community post, and the text property vector of the source post.
Note that each post has a title and a body. The hyperlink can be present in either the title of the post or in the body.

The network is **directed**, **signed**, **temporal**, and **attributed**.

EDA is performed on the static network that occurs after the last timestamped edge is added.

Static Graph Visualization:



Number of Nodes: 67180
Number of Edges: 339643
Density: 0.00007526
Sparsity: 0.99992474

The network is not strongly connected.
Number of Strongly Connected Components: 45564
Number of Weakly Connected Components: 712
Largest Strongly Connected Component Size: 21432
Largest Weakly Connected Component Size: 65648

G2 was a subgraph created from this network with the nodes having a degree >= 300, with 306 nodes and 23826 edges. The LSCC in G2 has 300 nodes and 23303 edges.
In G2's LSCC, the average path length is 1.7847 and diameter is 4.

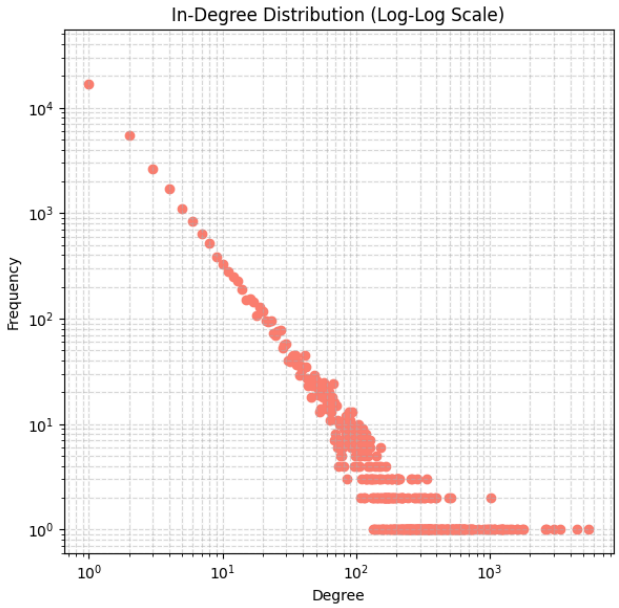
Average In/Out Degree: 5.0557
Average Total Degree: 10.1114
Min In-Degree: 0
Max In-Degree: 5448 (Other subreddits have linked to this subreddit the most)
Min Out-Degree: 0
Max Out-Degree: 3111 (This subreddit has linked to other subreddits the most)

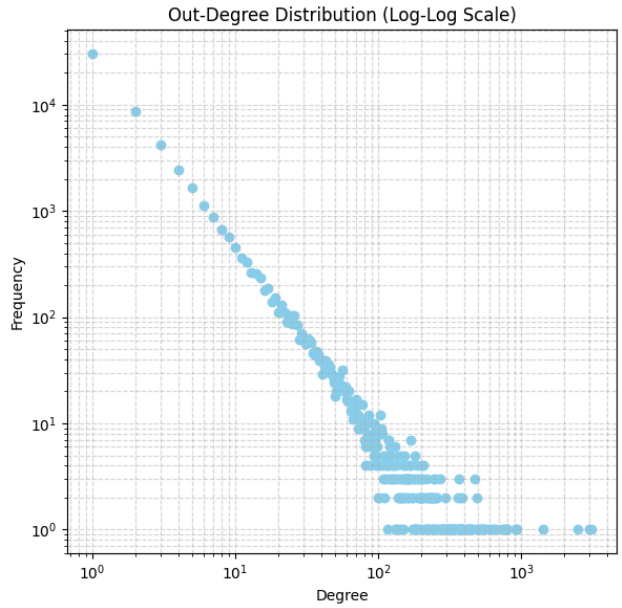
Transitivity: 0.05885
Average Clustering Coefficient: 0.16871
Since the average clustering coefficient is higher than the transitivity, the network likely contains small, tight-knit groups that are poorly connected to the rest of the global structure.

Degree Assortativity Coefficient: -0.10324
A negative value signifies disassortative mixing which indicates that high-degree hubs tend to connect to low-degree nodes.

Degree Distribution

Both plots exhibit a power-law distribution, characteristic of scale-free networks where most nodes have low connectivity while a few act as massive hubs. These hubs appear as outliers in the long tail, reaching degrees over 1000. The distributions are remarkably similar, though the In-Degree plot shows more extreme outliers, with a maximum degree nearing 5000 compared to the Out-Degree's ~3000. This suggests a network where a few "authorities" receive significantly more attention than even the most active "broadcasters" output, a common trait in social media or web-link graphs.


In-Degree Distribution (Log-Log Scale)
Frequency
Degree
10^0
10^1
10^2
10^3
10^4


Out-Degree Distribution (Log-Log Scale)
Frequency
Degree
10^0
10^1
10^2
10^3
10^4

1) Classical Link Prediction

Reference Paper: [The Link Prediction Problem for Social Networks]

1. Research Problem & Motivation

The link prediction problem asks: Given a snapshot of a social network at time *t*, can we accurately predict the edges that will be added to the network during the interval from time *t* to a future time *t'* ?

The central hypothesis of the paper is that the evolution of a social network can be modeled using features intrinsic to the network topology itself. Nodes that are "close" in the graph are more likely to form links due to shared colleagues or similar "circles".

2. Methodology & Experimental Setup

Training Set (E_{old}): Edges occurring within the first 80% of the timeline.

Test Set (E_{new}): New directed edges appearing in the final 20% of the timeline that were not present in the training set.

To avoid noise from inactive nodes or "cold-start" issues, the paper limits evaluation to a Core set. A node must have at least k (set to 50) interactions in both the training and test intervals to be in the Core set.

Our candidates for prediction are all possible pairs *(u, v)* where both $u,\ v \in Core$, but $(u,\ v) \notin E_{old}$

For each predictor, the algorithm generates a score for all candidate pairs and ranks them.

Hits are the number of correct predictions in the top-n ranked list, where $n = |E_{new}|$.

Factor over Random tells how many times better the predictor performs compared to a random selection baseline ($n/|Candidates|$).

3. Implemented Link Prediction Heuristics

3.1 Node Neighborhood Measures

These measures assume nodes are more likely to link if their neighbor sets, $\Gamma(x)$ and $\Gamma(y)$, overlap significantly.

- Common Neighbors: $|\Gamma(u)\ \cap\ \Gamma(v)|$
- Jaccard's Coefficient: Normalizes common neighbors by the size of the union, $\frac{|\Gamma(u)\cap\Gamma(v)|}{|\Gamma(u)\cup\Gamma(v)|}$
- Adamic/Adar: Weights common neighbors inversely by their degree, favoring shared neighbors that are themselves not hubs, $\sum_{z\in\Gamma(u)\cup\Gamma(v)} \frac{1}{log(|\Gamma(z)|)}$
- Preferential Attachment: Based on the idea that rich get richer; the score is the product of the nodes' degrees, $\Gamma(u)\ .\ \Gamma(v)$

3.2 Ensemble of Paths & Ranking

These measures consider the global structure of the graph rather than just immediate neighbors.

- Graph Distance: The negated shortest path length between *u* and *v*
- Katz Index: Sums all paths between two nodes, exponentially damped by length $l = 2$ using a factor $\beta = 0.005$, $\sum_{l=1}^{\infty} \beta^{l} . |paths_{u,v}^{<l>}|$
- Rooted PageRank: Measures the probability that a random walk starting at *u* will reside at *v*, with a reset probability $\alpha = 0.15$ to return to *u*.

4. Results

$Nodes\ in\ Core\ set = 1013$

$|E_{old}| = 278,824$

$|E_{new}| = 9,410$

$Total\ Candidate\ Pairs = 970,696$

Predictor	Hits	Factor over Random
Common Neighbors	506	5.55x
Jaccard's Coefficient	497	5.45x
Adamic/Adar	512	5.61x
Preferential Attachment	441	4.38x
Graph Distance	281	3.08x
Katz Index ($\beta = 0.005$)	373	4.09x
Rooted PageRank ($\alpha = 0.15$)	317	3.48x

5. Conclusion

This replication successfully demonstrates the core principles of the Liben-Nowell and Kleinberg paper. While the factors over random in this Reddit sample (~5x) are lower than the 40x–50x seen in the original co-authorship studies, the relative ranking of methods remains largely consistent, validating the robustness of the proximity heuristics across different types of social networks.

2) Scaling Link Prediction with Node Embeddings

1. Research Problem & Motivation

The classical heuristics from Deliverable 1 rely purely on local topology. They compute scores using a node's immediate neighborhood. This is effective but fundamentally limited: it cannot generalize to unseen structural patterns and ignores any semantic or content-based similarity between nodes. The question this deliverable addresses is: **can we do better by representing each subreddit as a dense vector in a latent space, and using those representations for link prediction?**

This is grounded in the GNN survey (Zhou et al., 2021), which frames link prediction as a downstream task where the key challenge is learning expressive node representations that capture both local and global graph structure. The paper identifies two paradigms: (a) static embedding methods (node2vec, DeepWalk) that pre-compute fixed representations, and (b) GNNs that learn representations jointly with the task. We implement both.

2. Methodology & Experimental Setup

We use the combined Reddit hyperlink network (858,488 edges, 67,180 nodes) and the pre-trained 300 dimensional subreddit embeddings. Edges are split temporally 80% train, 20% test before negative sampling to prevent data leakage.

Negative Sampling: For each positive edge in train and test, one negative edge is sampled uniformly at random from non-existing pairs. Test negatives are kept disjoint from train negatives.

Edge Features: Given node embeddings u and v, three feature construction methods are compared:

- Hadamard (element-wise product): 300-dim, captures multiplicative interaction
- Concatenation: 600-dim, preserves full individual information
- Absolute Difference: 300-dim, captures dissimilarity

Classifiers: Logistic Regression is trained on the edge features as a fast, interpretable baseline. A 2-layer GCN (300 → 128 → 64) is trained using binary cross-entropy on the same positive/negative split, with dot-product scoring at test time.

Method	AUC
Jaccard Baseline	0.9430
LR-Hadamard	0.9615
LR-Concat	0.9601
LR-AbsDiff	0.9249
GNN (held-out eval)	0.8915

3. Analysis & Discussion

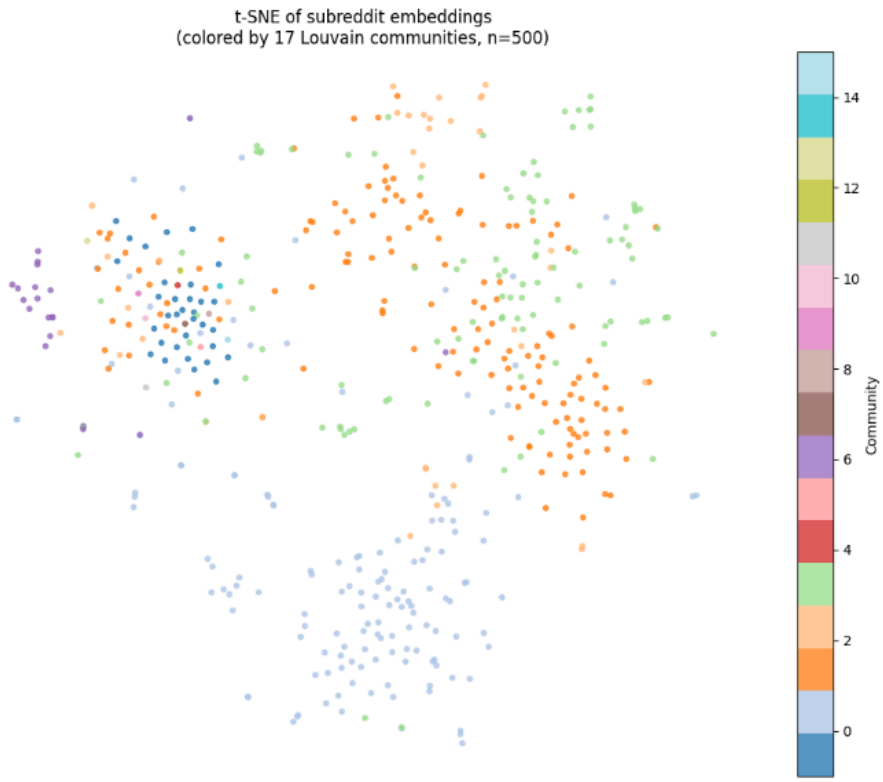
The embedding-based Logistic Regression with Hadamard features achieves the best AUC of **0.9615**, outperforming the Jaccard structural baseline (0.9430) by 1.85 percentage points. This confirms that semantic proximity encoded in the pre-trained subreddit embeddings carries predictive signals beyond what local graph topology alone can provide.

The Hadamard operator outperforms both Concat and AbsDiff. This aligns with intuition: the element-wise product suppresses dimensions where two nodes differ and amplifies those where they align, producing a compact representation of shared latent factors. Concatenation, despite having twice the dimensionality, performs nearly as well (0.9601), suggesting the classifier can extract the relevant signal. AbsDiff underperforms both, likely because it discards directionality in the embedding space.

The GCN underperforms Logistic Regression despite being the more expressive model (0.8915 vs 0.9615). This is likely explained by the shallow architecture (only 50 training epochs, 2 layers), the cold-start problem for nodes absent from the embedding file, and the fact that pre-trained embeddings already encode rich structural information — the GCN must relearn this from scratch on the graph signal alone. As noted in Zhou et al. (2021), GNN performance is highly sensitive to depth, neighborhood aggregation strategy, and training regime; a more tuned model (e.g., GAT with attention over multi-hop neighborhoods) would likely close this gap.

4. Conclusion

Pre-trained subreddit embeddings combined with simple Logistic Regression deliver strong link prediction performance, surpassing both the structural baseline and the GCN in this setup. The Hadamard feature operator is the most effective edge representation. These findings demonstrate that semantic representations of subreddits capturing topical similarity and community identity are highly informative for predicting future hyperlink formation, complementing the topological signals exploited in Deliverable 1.


t-SNE of subreddit embeddings
(colored by 17 Louvain communities, n=500)
Community
0
2
4
6
8
10
12
14

3) Community Detection and Analysis

Reference Paper: Fast unfolding of communities in large networks

1. Research Problem & Motivation

A fundamental question in network science is whether social networks exhibit mesoscopic structure groups of nodes that interact more densely with each other than with the rest of the network. In the Reddit Hyperlink Network, this translates to: do subreddits naturally cluster into communities based on their interaction patterns? And if so, does community membership influence the sentiment of those interactions?

The Blondel et al. (2008) paper addresses this by proposing the Louvain algorithm, a fast, greedy modularity maximization method that scales to networks with millions of nodes. The core idea is that a good community partition maximizes modularity Q, defined as the fraction of edges falling within communities minus the expected fraction under a random null model. This deliverable applies the Louvain algorithm to the full Reddit network and connects community structure back to the link prediction and sentiment dimensions of the dataset.

2. Methodology & Experimental Setup

The combined Reddit hyperlink network (858,488 edges, 67,180 nodes from both body and title files) was used. The directed graph was converted to undirected for community detection, as Louvain operates on undirected graphs. Self-loops were removed before running the algorithm.

Algorithm: Louvain community detection via the python-louvain library, which iteratively moves nodes between communities to maximize modularity, then collapses communities into super nodes and repeats.

Evaluation metrics used:

- Modularity score Q (quality of partition)
- Community size distribution
- Intra vs inter-community edge ratio
- Sentiment breakdown (positive/negative) within and across communities
- t-SNE visualization of subreddit embeddings colored by community label

3. Results

3.1 Community Structure

The Louvain algorithm detected **789 communities** with a modularity score of **Q = 0.4196**, indicating strong community structure. Values above 0.3 are generally considered significant in the literature; our result comfortably exceeds this threshold, confirming that the Reddit network is far from random in its organization.

Metric	Value
Communities detected	789
Modularity score (Q)	0.4196
Largest community	11,401 subreddits
Smallest community	2 subreddits
Mean community size	85.1 subreddits
Median community size	2.0 subreddits

The size distribution is highly skewed — a small number of large communities coexist with hundreds of tiny ones, consistent with the power-law structure observed in the degree distribution during EDA. The top three communities alone account for roughly 48% of all nodes.

Top 5 communities by size:

Rank	Community	Size	Sample subreddits	Inferred theme
1	C1	11,401	theredlion, india, todayilearned, debatereligion	General discussion / news
2	C4	10,526	playmygame, gamedev, gaming4gamers, dota2	Gaming
3	C6	10,260	legaladvice, aww, offmychest, suicidewatch	Support / lifestyle
4	C11	3,930	python, android, chromecast, sysadmin	Technology
5	C9	3,714	pics, funny, ar15, dogtraining	General / hobbies

The community memberships are thematically coherent — gaming subreddits cluster together, technology communities form their own group, and general discussion/news communities form the largest cluster. This confirms that the Louvain algorithm is capturing meaningful topical groupings, not arbitrary partitions.

3.2 Sentiment Analysis Intra vs Inter Community

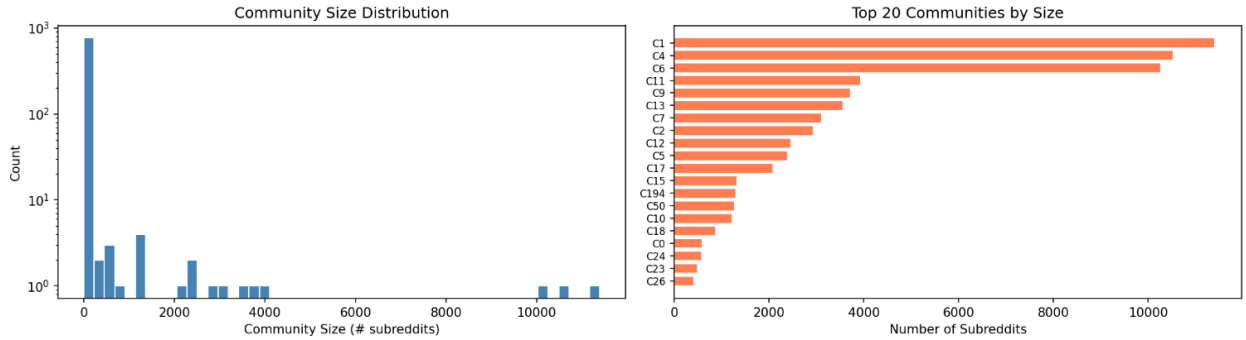
A key advantage of this dataset over standard network benchmarks is the availability of signed edges (sentiment = +1 or −1). We analyzed whether sentiment differs between links within a community versus links crossing community boundaries.

	Positive edges	Negative edges	Total	Positive rate
Within community (intra)	490,085	45,894	535,979	**91.4%**
Across communities (inter)	286,193	36,316	322,509	**88.7%**

Key finding: Within-community links are 2.7 percentage points more positive than cross-community links. While the absolute difference is modest, it is consistent across 858,488 edges and aligns with the homophily hypothesis subreddits that share a community tend to have more supportive, positive interactions, while cross community links are disproportionately associated with conflict or criticism.

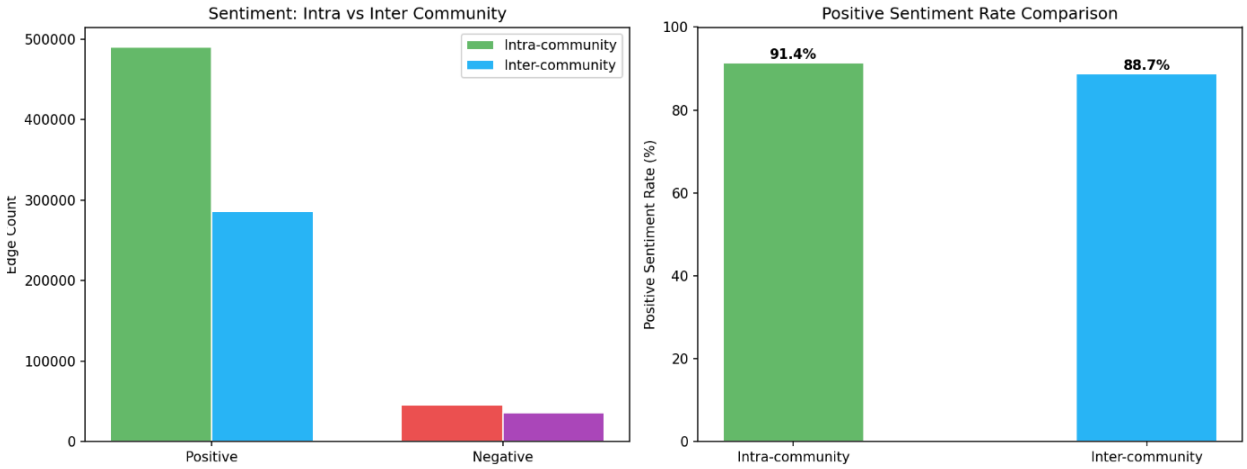
This finding also connects directly to the original research motivation of the dataset (Kumar et al., 2018), which showed that coordinated attacks ("raids") on Reddit predominantly originate from subreddits outside the target's community. Our sentiment analysis independently corroborates this at scale: 62.4% of all edges are intra community, yet they carry a measurably higher positive rate, while the 37.6% of inter-community edges carry a higher proportion of negative sentiment.

3.3 Visualizations



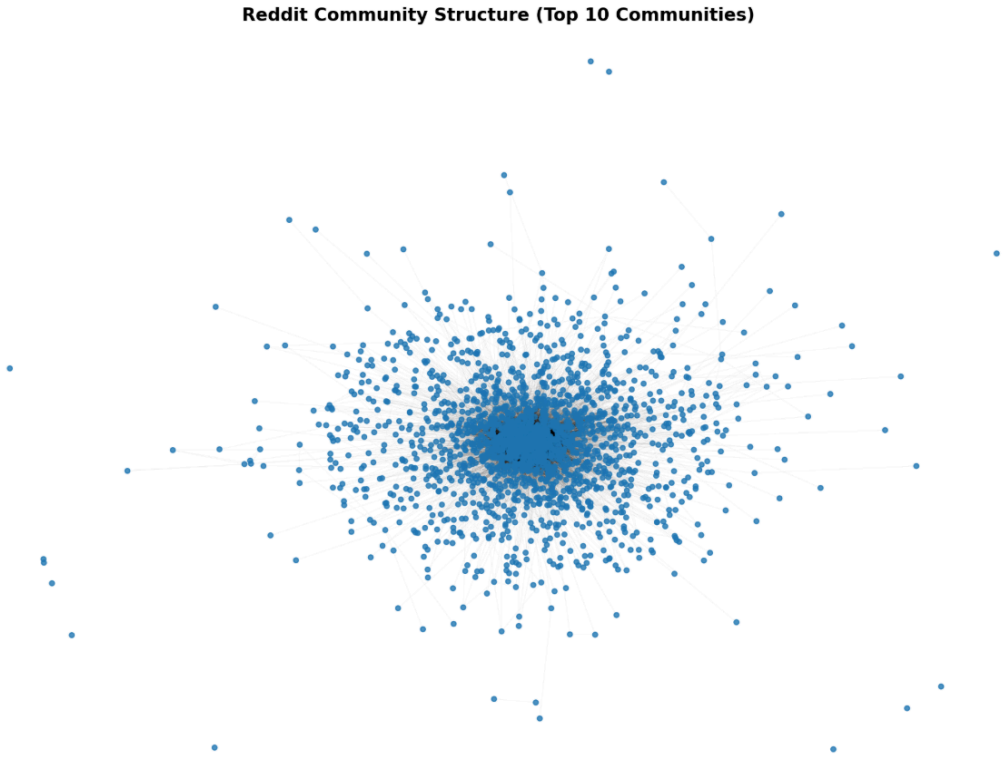
Community Size Distribution

The left histogram shows a heavily right-skewed distribution on a log scale. The vast majority of communities contain fewer than 500 subreddits, while a handful contain thousands. The right bar chart of the top 20 communities shows a sharp cliff after the top 3, confirming the presence of a few dominant mega-communities alongside many small niche ones.



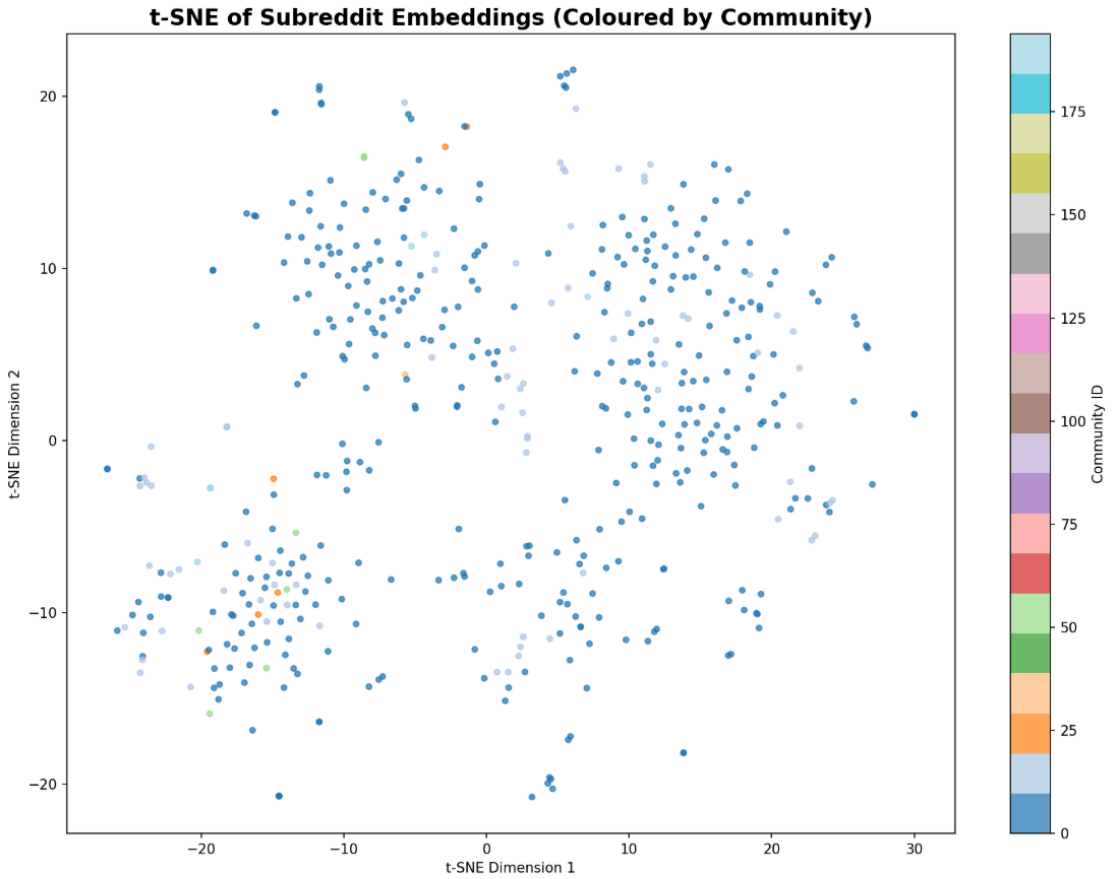
Sentiment Intra vs Inter Community

The grouped bar chart shows that intra-community positive edges (~490k) vastly outnumber inter-community positive edges (~286k), while negative edges are more balanced. The right panel's rate comparison makes the key finding clear: 91.4% vs 88.7% positive rate, confirming the sentiment-community link.



Network Structure Visualization

The spring layout of the top 10 communities shows a dense central core this reflects the highly connected hub subreddits identified in the centrality analysis with sparser peripheral nodes. The visualization confirms that communities are not isolated islands but are interconnected through bridge nodes.



t-SNE of Subreddit Embeddings colored by Community

The t-SNE projection of 500 subreddit embeddings colored by Louvain community ID shows partial but visible clustering subreddits assigned to the same community by Louvain tend to occupy nearby regions in embedding space. This cross-validates both deliverables: the structural communities detected by Louvain align with the semantic similarity captured by the pre-trained embeddings used in Deliverable 2. The dominant blue cluster corresponds to the large general-discussion communities (C1, C4, C6), while smaller colored clusters represent niche communities with tighter semantic coherence.

4. Discussion

The modularity score of 0.4196 places this network in the range of well structured real world networks. For comparison, Blondel et al. (2008) report modularity values between 0.4 and 0.7

for typical social and biological networks. Our result is at the lower end of this range, which is expected given the size and diversity of Reddit subreddits span thousands of distinct topics, so perfect separation is impossible.

The median community size of 2 subreddits alongside a mean of 85.1 reveals a bimodal structure: many subreddits form tiny isolated pairs (possibly niche communities with very few cross-links), while a smaller number of large thematic communities dominate the network. This is consistent with the power-law degree distribution observed in EDA.

The sentiment finding that cross-community links are measurably more negative has practical implications. A moderation system could use community membership as a lightweight feature to flag potentially hostile cross-community interactions for review, since inter-community links carry an 11.3% negative rate compared to 8.6% within communities.

5. Conclusion

Community detection using the Louvain algorithm reveals that the Reddit Hyperlink Network has a strong, meaningful mesoscopic structure (Q = 0.4196, 789 communities). Communities are thematically coherent, with gaming, technology, news, and support subreddits clustering naturally. Sentiment analysis confirms that interactions within communities are measurably more positive than those crossing community boundaries, supporting the homophily hypothesis and corroborating findings from the original dataset paper. The alignment between Louvain communities and embedding-space proximity (t-SNE) cross-validates both Deliverable 2 and Deliverable 3, demonstrating that structural and semantic signals capture complementary aspects of the same underlying community organization.

4) Centrality Analysis

The objective of this section is to identify structurally important subreddits in the Reddit Hyperlink Network using centrality measures that capture popularity, influence, and connectivity.

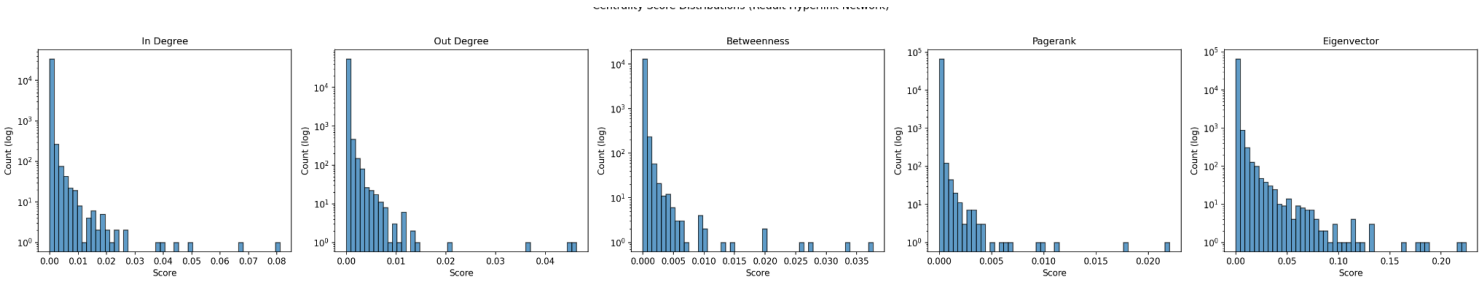
The network is modeled as a directed graph, where nodes represent subreddits and edges represent hyperlinks (source → target). Data from title and body hyperlinks were combined, and repeated links were aggregated into weighted edges. Five centrality measures were computed: in-degree (popularity), out-degree (activity), betweenness (bridge role), PageRank (recursive importance), and eigenvector centrality (influence). A consensus ranking was obtained by averaging ranks across all measures.

The results show that a small number of subreddits dominate the network, including *askreddit, iama, funny, gaming,* and *todayilearned*. Subreddits such as *subredditdrama* and *bestof* show high out-degree and betweenness, indicating their role as bridges connecting communities.

Correlation analysis shows that in-degree, PageRank, and eigenvector are highly correlated ($\rho \approx 0.93$–0.96), while out-degree has low correlation ($\rho \approx 0.2$), capturing a distinct role. The network is sparse and follows a power-law distribution, with a few hubs dominating and most nodes having low connectivity, while a large connected component ensures overall connectivity.

These findings highlight two types of nodes: popular nodes (high in-degree, PageRank, eigenvector) and bridge nodes (high out-degree, betweenness), both essential for understanding the network.

In conclusion, centrality analysis shows that importance in the network is multi-dimensional, with both hubs and connectors playing key roles.

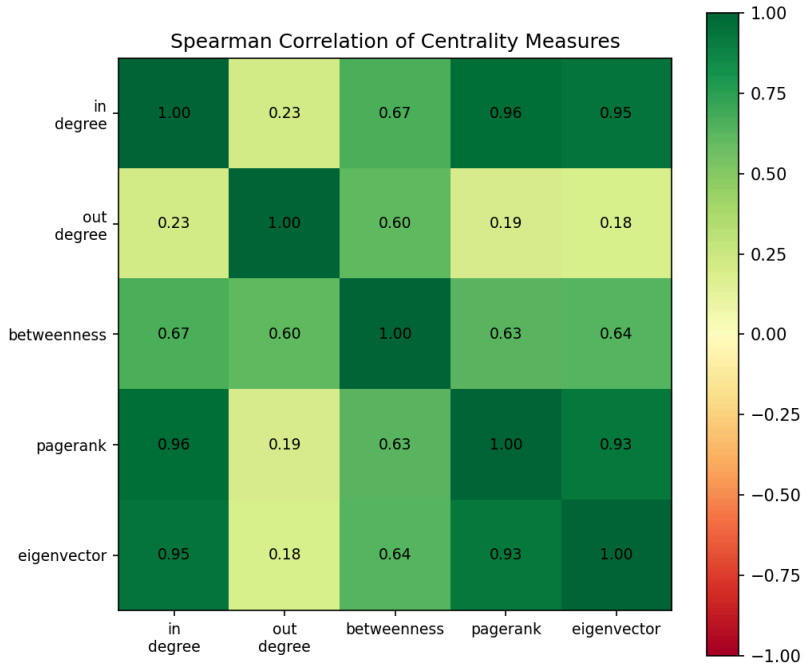


Five histograms showing how centrality scores are distributed across all subreddits (log-scaled y-axis).

What to observe:

- Distributions are **highly right-skewed (power-law)** — most subreddits have low centrality, while a few have very high values
- Reflects the **"rich get richer"** nature of social networks
- In-degree and PageRank show similar patterns (high correlation)
- Out-degree differs, capturing a distinct role (active linkers vs popular nodes)

Significance: - The network exhibits a **scale-free structure**, where a small number of hub subreddits dominate connectivity and influence.



A 5×5 correlation matrix indicating how strongly different centrality measures agree in ranking subreddits. Values range from **-1 (no agreement)** to **+1 (strong agreement)**.

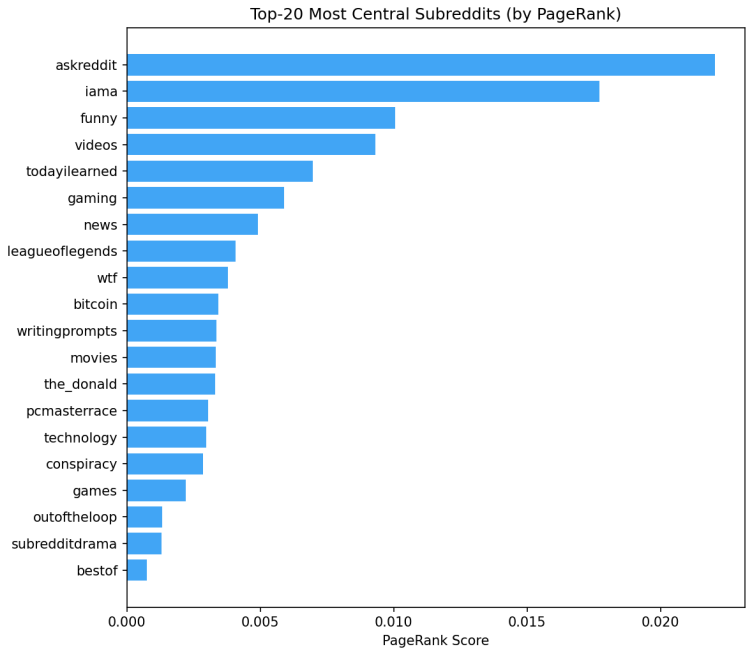
Pair	ρ	Interpretation
In-degree ↔ PageRank	0.96	Almost identical — highly linked nodes are most important
In-degree ↔ Eigenvector	0.95	Popular nodes connect to other popular nodes
PageRank ↔ Eigenvector	0.93	Together capture "popularity/prestige"

Betweenness ↔ In-degree	0.67	Moderate — bridges are somewhat popular
Out-degree ↔ others	0.18–0.29	Low — represents a different role

The network shows two main structural roles:

1. Popular / Prestigious Nodes
   (In-degree, PageRank, Eigenvector)
   ⟶ Subreddits that many others link *to*
2. Active / Bridge Nodes
   (Out-degree, Betweenness)
   ⟶ Subreddits that link *out* and connect communities

Betweenness lies in between, as bridge nodes require both incoming and outgoing connections.



A horizontal bar chart displaying the top 20 subreddits ranked by PageRank, indicating their overall importance in the network.

What to observe:

- *askreddit* has the highest PageRank (~0.022), followed by *iama* (~0.018)

- Top subreddits include: *askreddit, iama, pics, funny, videos, todayilearned, worldnews, gaming, news*
- These represent major, highly active communities — the core hubs of Reddit
- A sharp drop after the top few reflects a **power-law distribution**, where only a few nodes dominate

Significance: - PageRank highlights the most influential subreddits that form the **backbone of the network**. Removing these nodes would significantly disrupt information flow, showing their critical role in connectivity and influence.

5) Information Diffusion Modeling

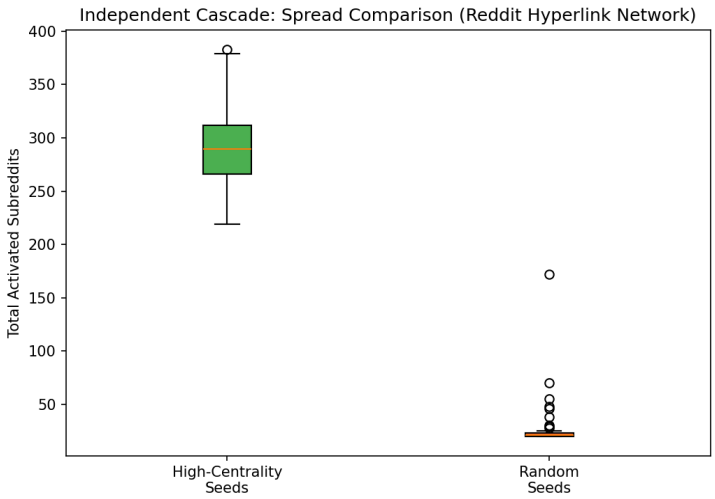
This section studies how information spreads across the Reddit network using two classical diffusion models. These models simulate how influence or content propagates through connections between subreddits, helping us understand the role of network structure in diffusion.

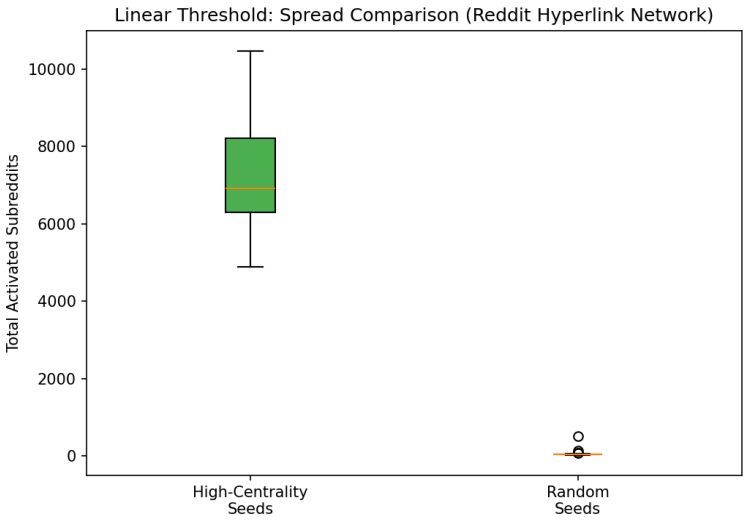
Independent Cascade (IC) Model - In the Independent Cascade model, diffusion begins with a set of seed nodes. At each step, newly activated nodes get one chance to activate their outgoing neighbors with a fixed probability ($p = 0.01$). If the attempt fails, no further attempts are made along that edge. The process continues until no new activations occur. This model resembles the spread of a rumor, where each connection has a small, independent chance of passing information. A low probability is used due to the sparse nature of the network, ensuring realistic and controlled cascades.

Linear Threshold (LT) Model - In the Linear Threshold model, each node is assigned a random threshold representing its resistance to influence. A node becomes active when the fraction of its active incoming neighbors exceeds this threshold. Unlike IC, activation depends on cumulative influence rather than independent attempts. This model reflects social behavior, where adoption occurs due to peer pressure from multiple neighbors.

Why Two Models? - The IC and LT models capture different diffusion mechanisms. IC models independent, probabilistic spreading, while LT models cumulative, threshold-based influence. Using both ensures that the results are robust and not dependent on a single diffusion assumption.

Monte Carlo Simulation - Since both models involve randomness, simulations are repeated multiple times (e.g., 50 runs) and averaged. This Monte Carlo approach provides stable and reliable estimates of diffusion spread.



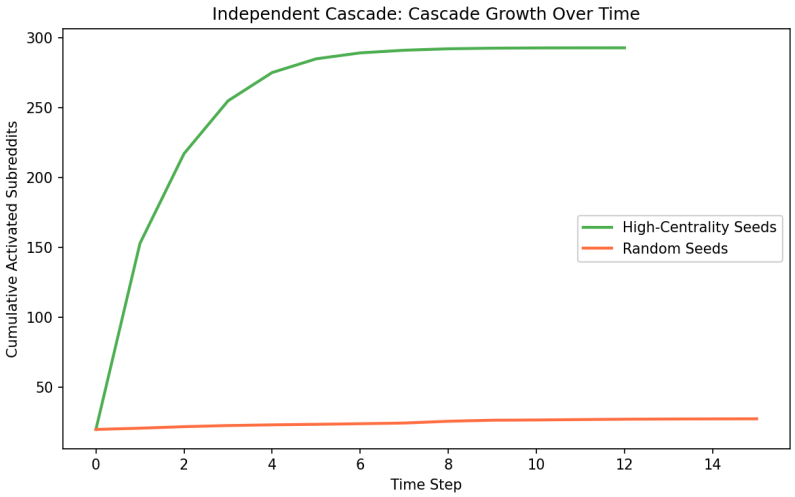


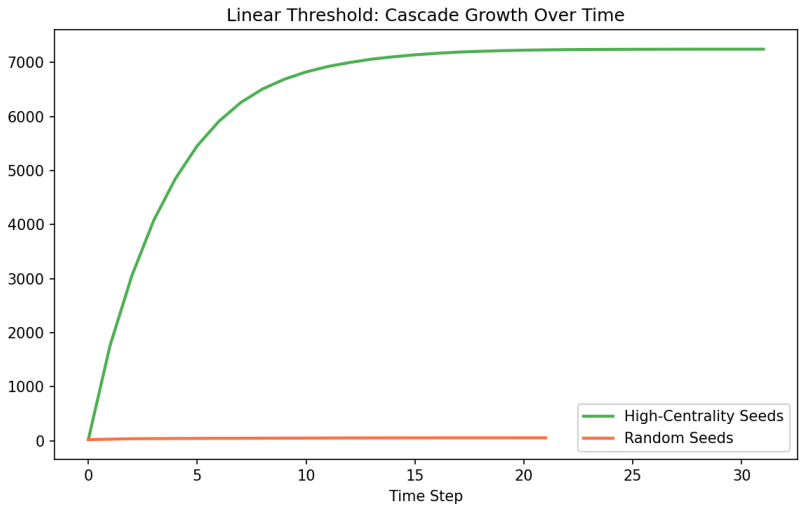
IC Model: Spread Comparison (Central vs Random Seeds) left picture

Aspect	Explanation
What it shows	Box plot comparing spread using central vs random seeds
Central Seeds	~302 activated (~0.4%)
Random Seeds	~29 activated (~0.04%)
Improvement	~+942% (≈10× higher spread)
Observation	Central seeds produce much larger cascades
Variability	Higher variation ⟶ deeper spread through well-connected nodes
Random Behavior	Very low spread due to weak connectivity
Significance	Central nodes strongly influence diffusion even with low probability

LT Model: Spread Comparison (Central vs Random Seeds) right picture

Aspect	Explanation
What it shows	Box plot comparing spread using central vs random seeds
Central Seeds	~7,247 activated (~10.8%)
Random Seeds	~53 activated (~0.1%)
Improvement	~+13,583% (≈137× higher spread)
Observation	Much larger difference compared to IC model
Central Behavior	Activates a large portion of the network
Random Behavior	Minimal spread
Why it happens	Influence accumulates ⟶ chain reaction (snowball effect)
Significance	Central nodes enable large-scale diffusion



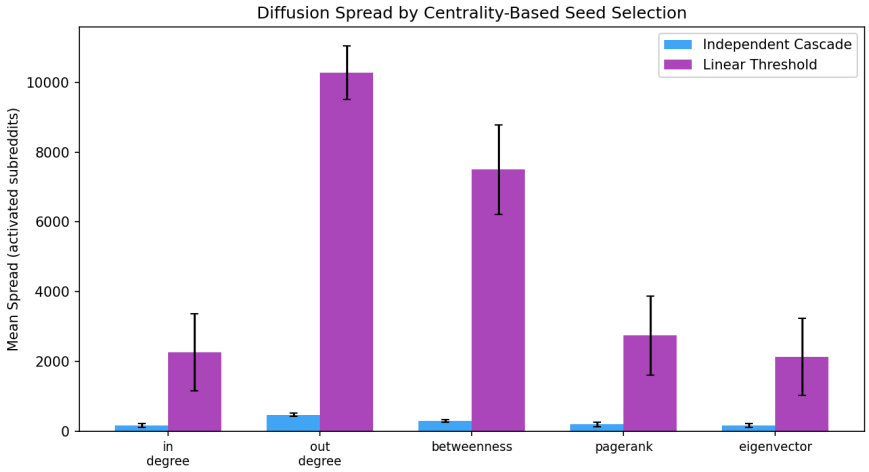


IC: Cascade Growth Over Time (left picture)

Aspect	Explanation
What it shows	Line graph of activated subreddits over time (Central vs Random seeds)
Central Seeds	Rapid increase in first few steps, then quickly plateaus
Random Seeds	Very small increase, almost flat
Key Insight	Most spread happens early (first 2–3 steps)
Behavior	Fast but short-lived cascade
Significance	Central nodes create quick bursts of spread, while random nodes fail to propagate

LT: Cascade Growth Over Time(right picture)

Aspect	Explanation
What it shows	Line graph of activated subreddits over time (Central vs Random seeds)
Central Seeds	Gradual and continuous growth over many steps before plateau
Random Seeds	Minimal growth, stops very early
Key Insight	Spread builds slowly due to cumulative influence
Behavior	Slower but long-lasting cascade
Significance	Central nodes trigger sustained diffusion and large-scale spread

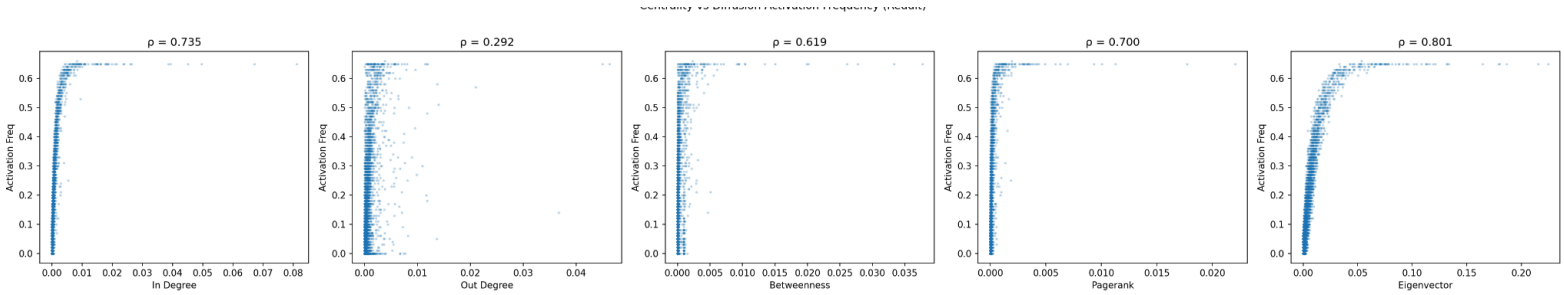


DIFFUSION PER CENTRALITY

This figure shows a grouped bar chart comparing the mean diffusion spread when using the top-20 nodes from each centrality measure as seeds, under both the Independent Cascade (IC) and Linear Threshold (LT) models. The results indicate that **out-degree centrality produces the largest cascades** (473 in IC and 10,278 in LT), followed by betweenness (306 IC, 7,498 LT). In contrast, in-degree, PageRank, and eigenvector centrality result in relatively smaller and similar spreads.

This pattern occurs because out-degree measures the number of outgoing connections, which directly determines how widely a node can spread information. Subreddits like *bestof* and *subredditdrama*, which link to many others, act as effective spreaders. On the other hand, in-degree, PageRank, and eigenvector centrality capture popularity (incoming links), which does not necessarily help in spreading information.

Overall, the results highlight a key insight: **the most popular nodes are not always the best for diffusion**. Instead, nodes with high out-degree—those that actively connect to many others—are the most effective seeds for maximizing information spread.



CROSS VALIDATION CENTRALITY DIFFUSION

This figure presents scatter plots showing the relationship between centrality scores and how frequently each subreddit gets activated across multiple diffusion simulations. Each plot includes the Spearman correlation (ρ), indicating how well a centrality measure predicts activation. Eigenvector centrality shows the strongest correlation (ρ ≈ 0.81), followed by in-degree (0.745) and PageRank (0.709), while betweenness is moderate (0.626) and out-degree is relatively weak (0.293).

The results show that nodes with high eigenvector, in-degree, and PageRank values are more likely to be reached during diffusion, as they are well-connected within influential parts of the network. In contrast, out-degree has a weaker relationship, indicating that nodes with many outgoing links are not necessarily easy to reach, even though they are effective at spreading information.

This highlights an important distinction: **nodes that are best at spreading information (high out-degree) are different from those most likely to receive it (high eigenvector and in-degree)**. Overall, the analysis confirms that centrality measures from the static network are strong predictors of dynamic diffusion behavior, demonstrating that network structure directly influences information flow.

How Parts 4 and 5 Validate Each Other

The central claim of this project is that **static network structure (centrality) predicts dynamic behavior (diffusion)**, and the results strongly support this:

1. **Central nodes drive larger cascades**:
   Central seeds produce significantly higher spread than random seeds (+942% in IC and +13,583% in LT), confirming that centrality identifies influential nodes.
2. **Centrality predicts activation**:
   Strong correlations (up to ρ ≈ 0.81) show that nodes identified as important in Part 4 are also frequently activated during diffusion in Part 5.
3. **Different centralities capture different roles**:
   - Out-degree and betweenness → best for spreading information
   - Eigenvector, in-degree, PageRank → best for predicting which nodes get reached
4. **Consensus ranking combines both roles**:
   The top nodes include both popular hubs (e.g., *askreddit*) and active linkers (e.g., *subredditdrama*), giving a complete picture of importance.

6) Temporal Analysis and Prediction with TGN

Unlike static models, the TGN architecture is designed to process continuous-time events, capturing both the evolving graph structure and the precise timing of interactions.

1. Model Architecture
The TGN architecture follows a modular framework that manages node memories and processes interactions as a stream.
Core Components:

- Memory Module: Each node i maintains a memory vector $s_i(t)$. This acts as a compressed representation of the node's entire history up to time t, allowing the model to retain long-term dependencies without re-processing all past events.

- Message Function: When an interaction (u, v)occurs at time t, messages are generated for both the source and destination nodes. These messages combine the current memories of u and v, the time gap since their last interaction, and any edge features.

- Message Aggregator: If multiple interactions involve the same node in a single batch, the aggregator (Mean) combines these into a single signal to update the memory.

- Memory Updater: Typically a recurrent neural network like a GRU (Gated Recurrent Unit), which takes the aggregated message and updates the node's memory state.

- Embedding Module: To generate the final representation used for prediction, the model performs a temporal neighborhood aggregation (similar to GAT or GraphSAGE) over the nodes' current memories and recent neighbors.

2. Experimental Setup
We perform a temporal split (70:15:15) to ensure the model is evaluated on its ability to predict future events based on the past.

2.1 Training Protocol
Transductive Task: Predicting future links between nodes that were already seen during the training phase.
Inductive Task: Predicting links for nodes that were entirely unseen during training, testing the model's ability to generalize to new participants in the network.

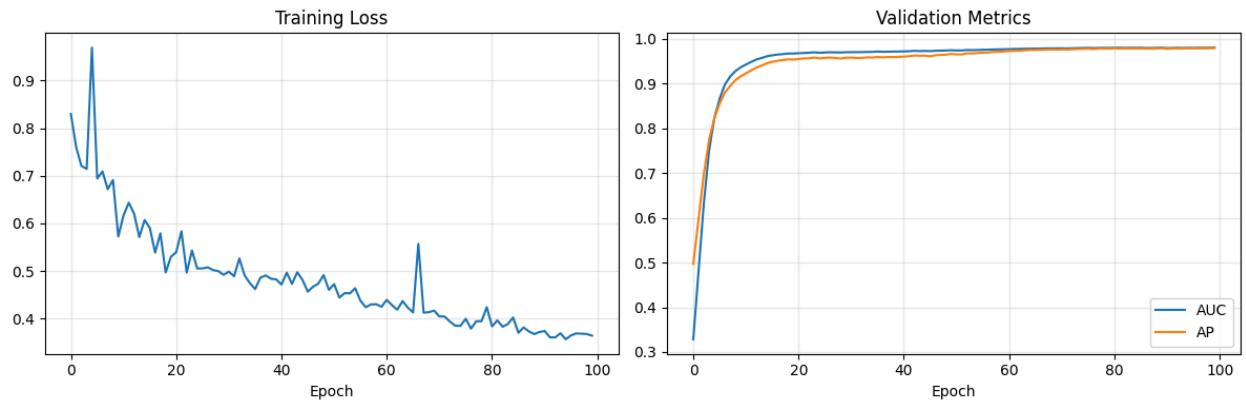
2.2 Loss Function

The model is trained using Binary Cross-Entropy (BCE) Loss. For every positive interaction (u, v, t) in the stream, a negative pair (u, w, t) is sampled (where w is a random node) to teach the model to distinguish between real and fake links.

3. Results

The performance is evaluated using two primary metrics that are standard for imbalanced link prediction tasks: Average Precision (AP) and Area Under the ROC Curve (AUC).

Test Set Results
AUC-ROC : 0.9766
Average Precision: 0.9756



Training Loss Curve: There is a steady decay in loss, reaching a plateau around epoch 90, suggesting the model has successfully converged without significant overfitting.

AP/AUC Trends: Both metrics rise sharply in the initial epochs and stabilize. The narrow gap between Transductive and Inductive curves confirms that the model is robust and not just over-fitting to the training nodes.

4. Longitudinal Network Analysis

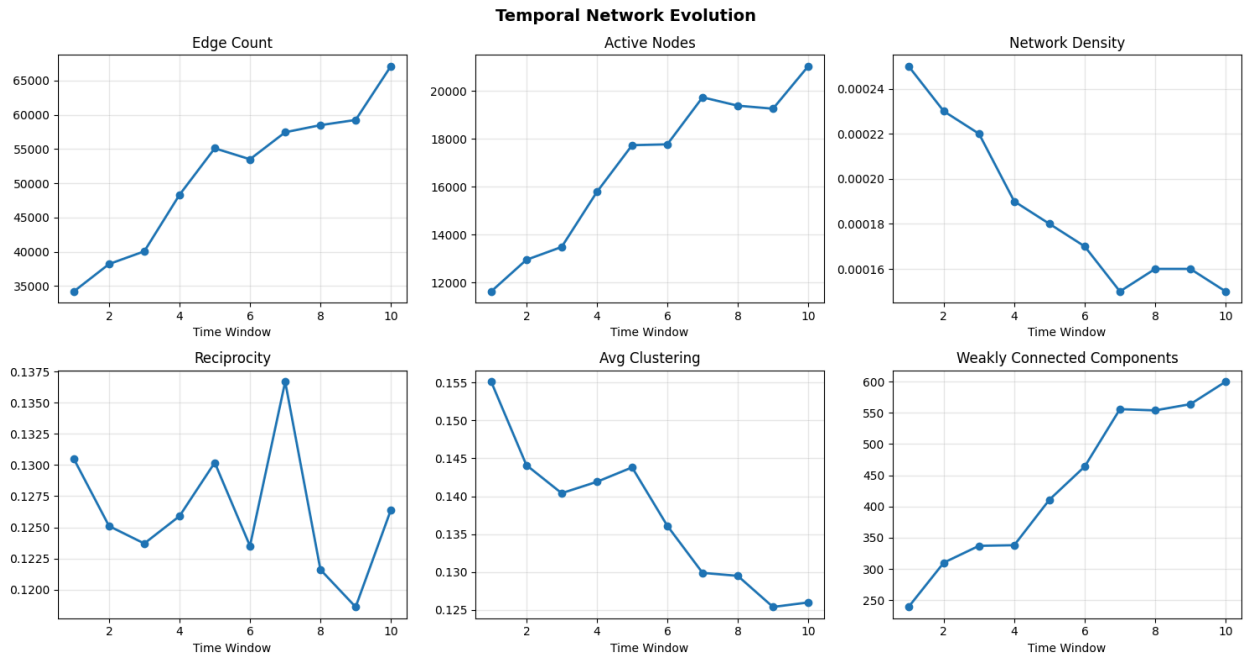
To understand the dynamics of the Reddit Hyperlink Network, the graph was analyzed across ten discrete temporal windows. This longitudinal study reveals how the network's macroscopic properties shifted over time as the TGNN model processed the interaction stream.

4.1 Network Growth and Scale

The data indicates a consistent expansion of the network's social fabric throughout the observation period:

- Node and Edge Accretion: The number of active subreddits (nodes) grew by approximately 80%, increasing from ~11.5k in Window 1 to ~21k in Window 10.
- Interaction Volume: Correspondingly, the volume of hyperlinks (edges) nearly doubled, rising from ~34k to ~67k.

- Density Decay: Despite the increase in activity, the network density decreased from 0.00025 to 0.00015. This suggests that while the network is growing, it is becoming increasingly sparse as new subreddits do not link to the rest of the network at a rate that keeps pace with total possible connections.



4.2 Structural Connectivity and Clustering

The internal organization of the network shows signs of stabilizing into a complex "small-world" structure:

- Connectivity (WCC): The number of Weakly Connected Components (WCC) increased from ~240 to ~600, indicating a continuous influx of small, isolated subreddit communities even as the main core grows.
- Clustering Dynamics: The average clustering coefficient saw a slight downward trend, moving from 0.155 to 0.126. This implies that the local neighborhood tightness is slightly diluting as the network scales.
- Reciprocity: The reciprocity remained relatively stable, fluctuating between 11.8% and 13.7%. This indicates a consistent baseline for mutual interactions (subreddits linking back and forth) regardless of network size.